Seeing the Process Through the Worker's Eyes

Event Logs from Egocentric Factory Videos

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The Problem

- Process mining needs **event logs** extracted from information systems
- **Manual shop-floor work** leaves few digital traces: picking a part, loading a blank into a die, carrying a tray
- At best, an MES logs order- or machine-level checkpoints; the **actual work** (sequence, duration, interruptions, handling overhead) stays **invisible**

The manual shop floor is a **blind spot** of process mining, yet a large share of **value creation** still happens there.

A New Data Source: Egocentric Video

- **Head-mounted cameras** record the shop floor from the **worker's perspective**
- Captures hand-object interaction, machine operation, movement **across stations**
- **Egocentric-10K**: thousands of hours of first-person factory footage, over 2,000 workers, real facilities

But **raw video is not event data**: no labels, no timestamps, no cases.

Frames from Egocentric-10K

The Goal

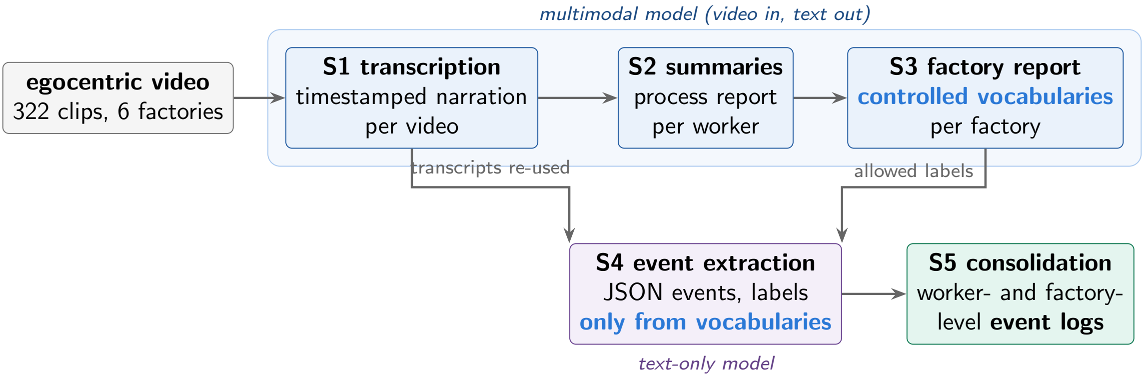
From raw footage to standard event logs, fully automated

A **five-stage pipeline** built only on **foundation models** turns egocentric factory videos into event logs: **no task-specific training, no human annotation**.

- **RQ1 (feasibility and validity)**: can such a pipeline produce **structurally valid** logs at the scale of hundreds of videos?
- **RQ2 (fidelity)**: how **faithfully** do the events reflect the observed work, temporally and semantically?
- **RQ3 (value)**: which **analyses** do the logs enable, and where do they hit the limits of the data?

The logs are **inferred observational data**, not ground truth.

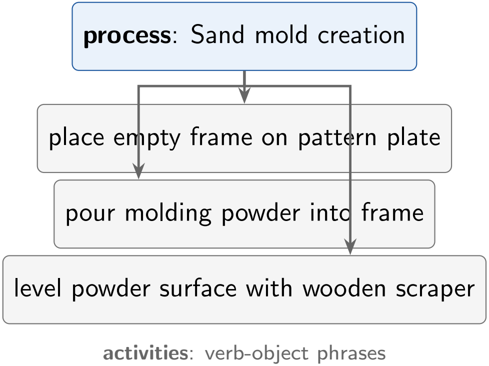
The Approach at a Glance



- **Late abstraction**: verbalize first, introduce activity labels only after a **factory-wide vocabulary** exists
- **Closed-world labeling**: extraction may only use vocabulary labels, which **suppresses hallucination** and label drift

Two-Level Controlled Vocabularies

- S3 induces, per factory, coarse **process labels** (12 to 24) and fine **activity labels** (55 to 131)
- Activities are **verb-object phrases**, human-readable and factory-specific
- The prompt **demands** coverage of non-value-adding behavior: transport, staging, waiting, setup, cleanup, inspection, handoffs
- Two labels per event = **two analysis granularities** from the same log



From Transcript to Events

S1 transcript entry [00:04:05 to 00:04:36]:
"A new empty frame is placed on the pattern plate. Powder is poured in . . . scrapes it across the surface."

constrained extraction:
closed vocabulary, transcript-anchored times

S4 event (one of several):
start **00:04:24**, end **00:04:36**, activity **level powder surface with wooden scraper**,
process **Sand mold creation**

- Transcript boundaries approximate **behavioral change points** and become candidate **event boundaries**
- Non-work footage is **filtered**; an empty event list is admissible
- Output is **schema-validated** JSON, checked against the vocabularies at parse time

The Derived Dataset

Factory	Domain	Workers	Videos	Events
001	metal stamping and bench assembly	11	97	2,288
002	garment finishing and packaging	10	54	1,541
003	investment-casting post-processing	11	72	2,449
004	garment sewing and confection	10	51	1,893
005	machining and electric motor assembly	11	22	1,103
006	sand-casting foundry	6	26	1,424
all		**59**	**322**	**10,698**

- Over **100 hours** of manual work, 13,304 transcript entries
- Synthetic time axis per worker (videos carry no wall-clock metadata): **durations and orderings** are meaningful, calendar dates are not
- Pipeline, prompts, logs, and assessment code **publicly released**

Evaluation: How

Logs are produced by **inference**, so validity comes first; the decisive question is **analytical value**:

- **Q1 validity preconditions**: schema completeness, vocabulary conformance, temporal coverage, boundary alignment with the transcript, and a manual semantic audit
- **Q2 analysis capability**: four classes on the released logs: **time composition** (A1), **fragmentation** of value-adding work (A2), **control-flow discovery** (A3), **worker benchmarking** (A4)
- **Q3 optimization evidence**: **triangulate** the pipeline's own qualitative reports against the log numbers: what do event logs add over narrative reporting?

All computations use pm4py on the released artifacts only.

Evaluation: Results

Q1: the logs are trustworthy

- **100%** vocabulary conformance (10,698 events)
- Median coverage ≈ **95%** for recordings up to 1 h (multi-hour recordings **truncate** and are excluded)
- **82%** of event boundaries within ±5 s of a transcript boundary
- Manual audit: precision **0.86**, recall ≈ **0.89**, zero contradicted labels

Q2, Q3: the logs add value

- Non-value-adding time varies **sevenfold** across factories (3.7 to 25.8%)
- Factory 003 spends **10.8%** of observed time **repairing defective castings**: a cost that points upstream to casting quality, visible **only in the log** (no written report on this factory mentioned it)
- Interruption profiles separate **material presentation** from **layout and logistics** problems
- Benchmarking localizes spreads up to **30 points** on identical tasks

What Can Be Improved: Data

- **Waiting is under-measured**: the extraction keeps only operational events, so idle stretches rarely become events at all; waiting-as-waste is invisible and every reported non-value-adding share is therefore a **lower bound**
- **Long recordings truncate**: extraction stops early on the seven multi-hour videos (median coverage 1.2%, against about 95% for clips up to one hour); **windowed extraction**, processing the transcript in chunks, would recover them
- **No product identifiers**: events say what the worker does, not **which piece** is worked on; linking events to objects (**object-centric** logs) would unlock pieces per hour and first-pass yield
- **Synthetic timestamps**: clips are concatenated on an artificial time axis, so durations and orderings are valid, but shifts, breaks, and utilization are not; recordings need **wall-clock anchoring**
- **One corpus**: six factories, all discrete manufacturing; the **analysis classes** transfer to other domains, the concrete numbers do not

What Can Be Improved: Pipeline

- **Vocabulary gaps become blind spots**. The induced vocabulary is the measurement instrument: what it does not name, the log cannot see. Factory 003 had **no transport label**, so transport was transcribed but could not be measured.
  Fix: make a standard set of categories (transport, waiting, rework, inspection) **mandatory** in vocabulary induction, so an absent behavior is a **measured zero**, not a blind spot

- **The video is never re-checked**. All fidelity checks compare events against the **transcript**, so errors made by the vision stage propagate silently into the logs.
  Fix: a **video-grounded audit** with several raters before any operational decision

- **Results depend on prompts and model versions**. Rerunning with newer foundation models may change the extracted logs.
  Mitigation: **versioned prompts** and released artifacts keep every analysis reproducible **without model access**

Takeaway

Foundation models turn **egocentric factory video** into **valid, analysis-ready event logs**, with **no training and no annotation**.

- 322 logs, 10,698 events, 6 real factories, over 100 hours of manual work (RQ1)
- Structurally sound and faithful within the dominant length regime: 100% vocabulary conformance, 82% boundary alignment, audit precision 0.86 (RQ2)
- The logs **quantify**, **refine**, and **extend** what qualitative reports say, and expose their own **vocabulary blind spots** (RQ3)

Pipeline, prompts, dataset, and assessment code:
`https://github.com/fit-alessandro-berti/annotated-egocentric-10k-dataset`

Appendix

Detailed Evaluation Results

All numbers from the released artifacts; recordings over one hour excluded from time statistics.

Appendix: Validity Preconditions (Q1)

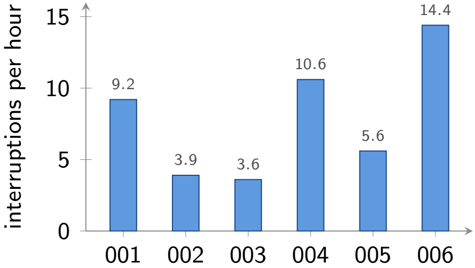
Factory	Workers	Videos	Events	Conform.	Coverage	Alignment
001	11	97	2,288	**100%**	94.8%	93.1%
002	10	54	1,541	**100%**	28.2%	71.8%
003	11	72	2,449	**100%**	93.0%	86.8%
004	10	51	1,893	**100%**	92.1%	90.4%
005	11	22	1,103	**100%**	25.0%	75.0%
006	6	26	1,424	**100%**	14.1%	62.1%
all	59	322	10,698	**100%**	52.8%	82.1%

- **Conformance**: share of event labels drawn from the factory vocabularies; **coverage**: share of the observed span covered by events; **alignment**: boundaries within ±5 s of a transcript boundary
- Low coverage in 002, 005, 006 stems from **multi-hour recordings** that truncate at extraction; recordings up to one hour reach a median coverage of about **95%**

Appendix: Time Composition and Fragmentation (A1, A2)

	VA	MH	TR	RW	Intr./h	VA run
001	93.1	4.1	1.8	0.0	9.2	4.6 min
002	96.3	0.1	2.3	0.0	3.9	8.8 min
003	82.1	0.4	0.0	**10.8**	3.6	9.0 min
004	89.9	3.0	3.5	0.0	10.6	3.6 min
005	80.6	5.4	9.7	0.0	5.6	6.2 min
006	74.2	**12.1**	7.8	0.0	**14.4**	**3.1 min**

VA = value-adding, MH = material handling, TR = transport, RW = rework (% of event-covered time); Intr./h = interruptions of VA work per hour; VA run = median uninterrupted VA stretch.



Factory **006** combines the highest interruption rate with the highest handling loss: a **layout and logistics** problem. Factory 001 is interrupted often but briefly: a **material presentation** problem.

Appendix: Worker Benchmarking (A4)

Workers on the **same dominant process** are directly comparable (at least 30 min of footage each):

Factory	Shared process	Workers	VA share	Mean VA run
001	manual mechanical assembly	6	89.7 to 97.8%	1.7 to 6.7 min
001	metal stamping	3	90.6 to 98.5%	3.2 to 7.9 min
003	surface finishing and polishing	5	**67.3 to 96.9%**	3.3 to 13.7 min
004	garment ironing	2	**52.8 to 77.1%**	3.5 to 4.1 min
004	overlock seaming	4	87.8 to 96.5%	1.5 to 10.3 min
005	manual lathe machining	2	**95.0 to 95.8%**	6.9 to 9.4 min

- Large spreads (**up to 30 points**) localize best practices worth standardizing, or hidden structural differences between stations
- Near-identical machinists in 005 show the opposite: **station design**, not behavior, drives that factory's losses
- Spreads are **hypotheses** about stations, not verdicts on individuals

Appendix: Triangulation Against Qualitative Reports (Q3)

Report claim (qualitative)	Log evidence (quantitative)	Relation
001: bottleneck is the lack of dedicated material handlers	14.6 handling episodes/h, mean 15 s, only 5.9% of time	**Refined**: presentation, not headcount
006: excessive manual handling, facility-wide bottleneck	19.8% of time in handling and transport, 28.4 episodes/h	**Confirmed and quantified**
003: workers halt work to carry heavy trays	transport share 0.4%: the vocabulary has **no transport label**	**Blind spot exposed**
003: report silent on rework	defect patching = 10.8% of observed time	**New, log-only finding**

Event logs **quantify** claims for prioritization, **refine** the intervention, surface **new findings**, and expose what the induced vocabulary **cannot see**.